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Producers and consumers

Lindeman's research showed how an ecosystem is powered by a stream of energy from one organism to another. The organisms can be grouped into discrete "trophic levels" (feeding levels)—from producers (plants and algae), which absorb energy in the form of sunlight to make food, to consumers (animals). "Primary consumers" are the herbivores that eat the plants; "secondary consumers" are animals that eat the herbivores. Each trophic level depends on the preceding one for its survival. At the same time, dead material accumulating from each stage is broken down by decomposers, such as bacteria and fungi, and materials in the form of nutrients are recycled back to feed plants and algae.

Lindeman also demonstrated how some of the energy at each trophic level is lost as waste, or converted into heat when organisms respire. By combining the results of his own study with data from a wide range of other sources, he was able to build a picture of this system as it worked in Cedar Creek Bog.

British ecologist G. Evelyn Hutchinson, considered to be one of the founding fathers of modern ecology, was Lindeman's mentor at Yale University. He recognized the importance of his student's work to the future development of ecology, and he lobbied for Lindeman's paper to be accepted. Lindeman, who had always suffered from ill health, died in 1942 from cirrhosis of the liver at the tragically young age of 27, just four months before his trophic-dynamic paper—now seen as a classic in its field—was finally published.